An Adaptive Multi-Agent System for Student Competency Analysis Across Cognitive Domains

1st Author Name
Dept. of Computer Science
Institution Name
City, Country
email@institution.edu

2nd Author Name
Dept. of Computer Science
Institution Name
City, Country
email@institution.edu

3rd Author Name
Dept. of Computer Science
Institution Name
City, Country
email@institution.edu

***Abstract*—Traditional educational assessments fail to capture the true ability of learners by presenting identical questions regardless of individual competency levels. This paper presents an Adaptive Multi-Agent MCQ Difficulty Prediction System that dynamically evaluates student responses and adjusts question difficulty in real time using a fully LLM-driven pipeline. Building upon the multi-agent competency-building framework of Outay *et al.* [1], we adapt the Assess–Predict–Oversee–Transit (APOT) model to a question-based adaptive testing context where all core decision-making is handled by intelligent LLM agents rather than deterministic formulas. The system is orchestrated through the CrewAI multi-agent framework, deploying three specialized agents powered by Google Gemini 2.5 Flash: an MCQ Evaluator Agent, a Difficulty Predictor Agent, and an Adaptive MCQ Overseer Agent. Each agent reasons over student answer correctness and response time to collaboratively determine the difficulty level of the current question and recommend the most appropriate difficulty for the next question. This design eliminates reliance on pre-calibrated Item Response Theory models and static threshold formulas, instead leveraging the contextual reasoning capabilities of large language models to produce transparent, flexible, and explainable adaptive assessments. A design-level analysis demonstrates the logical consistency of the agent collaboration pipeline, the correctness of the adaptive difficulty rules embedded in agent prompts, and the qualitative responsiveness of the system to diverse student performance patterns.**

***Index Terms*—adaptive assessment, MCQ difficulty prediction, multi-agent system, CrewAI, Gemini, LLM-based adaptation, competency analysis, personalized assessment**

I. Introduction

Traditional educational assessments adopt a one-size-fits-all model, presenting identical questions to all students regardless of individual competency. This approach inherently assumes uniform learning ability and pace among learners, which is rarely the case in real-world educational environments. As a result, high-performing students may find the assessment too easy and unchallenging, while low-performing students may struggle excessively, leading to frustration and disengagement. Consequently, such systems fail to capture the true ability of learners at either extreme of the performance spectrum and provide little actionable insight for instructors regarding individual strengths, weaknesses, and learning trajectories [2].

Furthermore, most existing assessment systems provide limited or generic feedback that does not help students understand their specific areas of strength and weakness. This lack of personalization hinders accurate competency measurement, reduces student engagement, and limits learning efficiency [9]. Without continuous adaptation based on student performance, assessments remain rigid and unable to reflect the dynamic nature of learning [10]. There is therefore a pressing need for an adaptive assessment system that adjusts to each student's abilities in real-time and provides data-driven, contextually intelligent feedback to enhance the learning process.

Computerized Adaptive Testing (CAT) has emerged as a promising paradigm to address these limitations by dynamically adjusting question difficulty based on ongoing student responses [3]. However, most CAT systems rely on Item Response Theory (IRT) models, which require large-scale pre-calibrated item banks and statistically validated parameters for each question [3]. This dependency significantly limits their applicability in smaller educational settings where such datasets are unavailable. Additionally, IRT-based systems often lack transparency, making it difficult for educators to interpret how and why certain questions are selected, thereby reducing trust in formative educational contexts [4].

To address these challenges without imposing rigid mathematical formulas, recent advances in Large Language Models (LLMs) offer an alternative pathway: encoding adaptation logic directly into the reasoning capabilities of intelligent agents. Rather than computing a numerical Learning Index or calibrating difficulty via statistical thresholds, LLM-powered agents can reason over correctness and response time holistically, applying natural language adaptation rules to produce context-aware difficulty decisions.

Outay *et al.* [1] proposed a recursive APOT model managed by a Multi-Agent System (MAS) for competency building, introducing components such as the Student Online Learning Potential Assessment (SoLPA) and Student Online Learning Trajectory Algorithm (SoLTA) for tracking learner progress. While their work provides a strong foundation for competency-based learning, their framework primarily focuses on augmented reality (AR)-based vocational training environments involving guided instruction, and relies on quantitative KPI metrics and deterministic indices. Our system adapts the spirit of the APOT model—its cyclic Assess, Predict, Oversee, Transit stages—into a fully LLM-driven, question-based adaptive testing environment.

This paper proposes an Adaptive Multi-Agent MCQ Difficulty Prediction System with the following key contributions:

1) A three-agent CrewAI pipeline powered by Google Gemini 2.5 Flash, replacing formula-driven scoring with LLM-native contextual reasoning.
2) An MCQ Evaluator Agent that assesses answer correctness and provides a natural language explanation.
3) A Difficulty Predictor Agent that infers question difficulty from correctness and response time using psychometric reasoning encoded in agent prompts.
4) An Adaptive MCQ Overseer Agent that synthesizes prior agent outputs to produce both a current difficulty classification and a recommended next-question difficulty, guided by explicit adaptive rules.
5) An interpretable, prompt-engineered adaptation mechanism that eliminates IRT pre-calibration while maintaining responsiveness and explainability.

II. Related Work

A. Multi-Agent Systems in Education

Multi-Agent Systems (MAS) have gained significant attention in educational technology due to their ability to model complex, distributed, and intelligent behaviors [7]. Outay *et al.* [1] proposed the APOT model, which leverages a MAS architecture to manage different stages of competency development. Their system includes agents such as the Learning Style Assessment Agent (LSAA), SoLPA, Student Online Learning Outcome Assessment (SoLOA), and SoLTA, each responsible for specific aspects of the learning process. These agents collaboratively compute a Learning Index (LI) and Competency Index (CI) to monitor learner progress across sessions. However, their framework targets vocational training in AR environments, limiting its direct applicability to digital, question-based assessment contexts [1]. Our work adapts the APOT's multi-agent collaboration philosophy to a question-driven setting, replacing quantitative KPIs with LLM-based reasoning over correctness, time, and difficulty trajectory.

B. Computerized Adaptive Testing

Computerized Adaptive Testing (CAT) has been extensively studied since the 1970s and is widely regarded as a robust method for personalized assessment [2]. CAT systems typically rely on IRT, where each question is associated with parameters such as difficulty, discrimination, and guessing probability [3]. While statistically rigorous, IRT-based CAT systems require large, well-calibrated item banks that are expensive and time-consuming to develop [4]. The probabilistic nature of IRT models also makes them difficult to interpret for educators. Our approach replaces IRT with LLM-based agents that reason over observable performance signals—correctness and time—using transparent natural language rules, offering ease of deployment without sacrificing adaptive responsiveness.

C. LLM-Augmented Assessment Frameworks

Recent advances in Large Language Models have enabled their integration into educational systems for tasks such as feedback generation, tutoring, and content creation [11]. Frameworks such as CrewAI [5] and AutoGen facilitate the deployment of LLM-augmented multi-agent systems where agents collaborate to perform complex reasoning tasks. Bloom's taxonomy [8] provides a cognitive hierarchy that informs how question difficulty scales with learner ability; our three-level difficulty scheme (Easy, Medium, Hard) is grounded in this framework. Unlike prior hybrid systems that restrict LLMs to feedback generation only, our system delegates the full assessment pipeline—including answer evaluation, difficulty prediction, and adaptation decision—to LLM agents, demonstrating that prompt-engineered agents can reliably implement psychometric adaptation logic without deterministic formulas.

III. System Architecture

A. Assessment Workflow

The adaptive MCQ difficulty prediction system follows a per-question adaptive loop. For each question presented to the student, the system captures the student's selected answer and the time taken to respond. These inputs are passed to the CrewAI pipeline, which executes three agents sequentially. The MCQ Evaluator Agent first determines whether the answer is correct or incorrect. The Difficulty Predictor Agent then classifies the difficulty of the current question based on correctness and time. The Adaptive MCQ Overseer Agent synthesizes these results to output both the current question's difficulty classification and the recommended difficulty level for the next question. This loop repeats for each question in the session, enabling real-time adaptation of assessment difficulty.

B. Architecture Overview

The system is organized into three interconnected layers, each with clearly defined responsibilities.

The **Presentation Layer** handles question display, answer input collection, and timer visualization. The **Orchestration Layer** coordinates the sequential execution of the three CrewAI agents using the CrewAI framework with Google Gemini 2.5 Flash as the underlying LLM, managing data flow and task dependencies across the pipeline. The **Analytics Layer** is the core intelligence of the system, where each agent applies its specialized role—evaluation, difficulty prediction, and adaptive decision-making—through LLM reasoning over structured input data.

The original APOT learning model [1] follows a cyclic process of Assess, Predict, Oversee, and Transit stages. In our adaptation, the *Assess* stage maps to the MCQ Evaluator Agent determining correctness; the *Predict* stage maps to the Difficulty Predictor Agent classifying question difficulty; the *Oversee* stage maps to the Adaptive MCQ Overseer Agent reviewing the combined analysis; and the *Transit* stage maps to the Overseer Agent's final recommendation of the next question's difficulty level.

IV. Multi-Agent Framework

The multi-agent architecture distributes the adaptive assessment task across three specialized agents, each defined with a distinct role, goal, and backstory in the CrewAI framework. All agents are powered by Google Gemini 2.5 Flash via the CrewAI LLM interface with a temperature of 0.7. The agents execute sequentially with context passed between tasks, ensuring that each agent builds upon the output of its predecessor.

A. MCQ Evaluator Agent

The **MCQ Evaluator Agent** serves as the first stage of the pipeline and corresponds to the *Assess* phase of the APOT model. Its role is strictly defined as a fair and accurate MCQ examiner: given the question text, the correct answer, and the student's selected answer, it determines whether the response is correct or incorrect and provides a one-line explanation. The agent is configured with `allow_delegation=False` to ensure it does not offload this factual judgment to other agents.

The agent's task receives a structured input block containing:

- The question text
- The correct answer
- The student's selected answer
- The time taken in seconds

Its expected output is a binary correctness verdict with a brief justification, providing the factual foundation for subsequent agents.

B. Difficulty Predictor Agent

The **Difficulty Predictor Agent** corresponds to the *Predict* phase of the APOT model. Its goal is to predict the true difficulty of the current question—classified as Easy, Medium, or Hard—based on the correctness verdict from the Evaluator Agent and the response time. The agent's backstory encodes psychometric reasoning: a fast and correct response indicates an Easy question; a slow but correct response indicates a Medium question; an incorrect response indicates a Hard question regardless of time.

This design reflects the insight, consistent with adaptive testing literature [2], that a student's response time in conjunction with correctness serves as a behavioral proxy for item difficulty. Rather than computing a numeric difficulty parameter as in IRT [3], the agent applies this reasoning in natural language, producing an interpretable and auditable prediction. The agent outputs the predicted difficulty level along with its reasoning.

C. Adaptive MCQ Overseer Agent

The **Adaptive MCQ Overseer Agent** is the most complex agent in the pipeline and serves both the *Oversee* and *Transit* functions of the APOT model. It receives the outputs of both prior agents as context and synthesizes them to produce two key decisions: (1) the confirmed difficulty classification of the current question, and (2) the recommended difficulty level for the next question.

The agent's backstory encodes an explicit adaptive rule set grounded in the progressive challenge principle [9] and Bloom's taxonomy [8]:

If the answer is correct:

- Easy → Next: Medium (increase challenge)
- Medium → Next: Hard (advance to next level)
- Hard → Next: Hard or Medium (maintain mastery)

If the answer is incorrect:

- Hard → Next: Medium (step down one level)
- Medium → Next: Easy (step down one level)
- Easy → Next: Easy (stay at same level for confidence building)

The agent is configured with `allow_delegation=True`, enabling it to consult prior agents if additional clarification is needed. Its output follows a structured natural language format:

```
Questions Difficulty: [Easy/Medium/Hard]
Next Question Difficulty: [Easy/Medium/Hard]
Reasoning: [explanation]
Recommendation: [additional actions]
```

This structured prompt output is then parsed programmatically to extract the difficulty labels for system use.

D. CrewAI Orchestration

The three agents are assembled into a `Crew` with sequential process execution, `memory=False` for stateless per-question operation, and `max_rpm=10` for API rate compliance. Three corresponding `Task` objects define the inputs and expected outputs for each agent, with Task 3 explicitly receiving Task 1 and Task 2 as context. The `crew.kickoff()` call executes the full pipeline and returns the Overseer Agent's final output, which is parsed to extract the current and next difficulty values.

Fig. 1 illustrates the three-agent sequential pipeline for a single MCQ response.

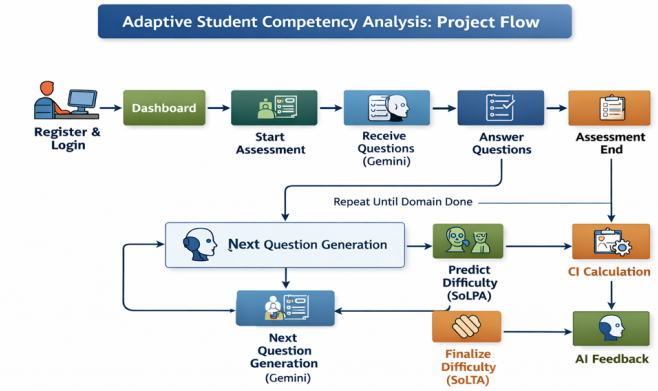

Fig. 1. Three-agent CrewAI pipeline for adaptive MCQ difficulty prediction. Each question response triggers sequential execution of the Evaluator, Predictor, and Overseer agents powered by Gemini 2.5 Flash.

V. Methodology

The methodology focuses on the agent prompt design, the adaptive rule encoding, and the information flow that enables the system to make coherent difficulty predictions without deterministic formulas.

A. Input Representation

For each MCQ response, the system constructs a structured input string passed to the Evaluator Agent's task:

```
Question: <question text>
Correct Answer: <correct option>
Student Selected Answer: <student option>
Time Taken by Student: <seconds> seconds
```

This input captures all observable signals available for a single question interaction: factual correctness (determined by comparing answers) and response efficiency (captured by time taken). Both signals together constitute the behavioral fingerprint that the pipeline reasons over.

B. Agent Role and Goal Specification

Each agent's behavior in CrewAI is governed by three natural language fields: `role`, `goal`, and `backstory`. These fields function as the system prompt for the underlying Gemini model, shaping its reasoning approach for each task. The role defines the agent's identity within the pipeline, the goal specifies its objective, and the backstory provides the contextual knowledge and heuristics the agent should apply.

For the Difficulty Predictor Agent, the backstory explicitly encodes the time-correctness heuristic: "Fast + Correct = Easy question. Slow + Correct = Medium. Wrong = Hard." This concise rule set mirrors the psychometric intuition behind time-aware adaptive testing [2] and eliminates the need for IRT parameter estimation.

For the Adaptive MCQ Overseer Agent, the backstory encodes the full bidirectional transition table (increase on success, decrease on failure) aligned with the Zone of Proximal Development concept [9], ensuring the system keeps students within their optimal challenge zone.

C. Sequential Task Execution and Context Passing

The sequential process in CrewAI ensures tasks execute in order with outputs from earlier tasks available as context for later ones. Task 3 (Overseer) explicitly declares `context=[task1, task2]`, giving the Overseer Agent access to both the correctness verdict and the difficulty prediction before making its final decision. This mirrors the information aggregation role of the Overseer stage in the APOT model [1], where cumulative performance data informs the transition decision.

D. Output Parsing

The Overseer Agent's output is parsed line by line to extract the structured difficulty labels. Lines containing the keywords `Questions Difficulty:`, `Next Question Difficulty:`, `Reasoning:`, and `Recommendation:` are individually parsed to populate a result dictionary. Default values of "Medium" are used as fallbacks if parsing fails, ensuring the system degrades gracefully under LLM output variability. The full raw output is also retained in the result dictionary as `full_analysis` for logging and audit purposes.

VI. System Analysis and Discussion

This section presents a design-level analysis examining the logical consistency of the agent roles, the adaptive rule coverage, and the system's behavior across representative student performance patterns.

A. Agent Role Separation and Pipeline Integrity

The three-agent design enforces a separation of concerns that mirrors the APOT model's staged reasoning. The Evaluator Agent handles only factual comparison, ensuring correctness determination is isolated from any difficulty reasoning. The Predictor Agent applies psychometric heuristics exclusively to the current question, without considering adaptation goals. The Overseer Agent alone integrates correctness, difficulty prediction, and adaptive rules to produce actionable recommendations. This staged design prevents conflation of concerns and makes each agent's reasoning independently auditable.

Table I summarizes the mapping between APOT phases and the three agents.

TABLE I
APOT Phase Mapping to CrewAI Agents

APOT Phase	Agent	Function
Assess	MCQ Evaluator	Answer correctness verdict
Predict	Difficulty Predictor	Current question difficulty classification
Oversee	Adaptive Overseer	Synthesis and current difficulty confirmation
Transit	Adaptive Overseer	Next question difficulty recommendation

B. Adaptive Rule Coverage Analysis

Table II enumerates all six possible correctness-difficulty combinations handled by the Overseer Agent, verifying complete coverage of the transition space.

The rule set covers all six combinations in the $\{\text{Correct}, \text{Incorrect}\} \times \{\text{Easy}, \text{Medium}, \text{Hard}\}$ space, ensuring no undefined transitions exist. The design prevents two failure modes common in adaptive systems: (1) oscillation, where a student alternates between difficulty levels without convergence, is mitigated by the "maintain mastery" Hard+Correct rule; and (2) indefinite floor trapping, where a student is stuck at the lowest level, is mitigated by the Easy+Correct escalation to Medium.

TABLE II
ADAPTIVE DIFFICULTY TRANSITION RULES ENCODED IN OVERSEER AGENT

Correct?	Current	Next	Rationale
Yes	Easy	Medium	Progressive challenge
Yes	Medium	Hard	Advance to next level
Yes	Hard	Hard	Maintain mastery
No	Hard	Medium	Step down one level
No	Medium	Easy	Step down one level
No	Easy	Easy	Reinforce confidence

C. Behavioral Trace Across Performance Profiles

Table III illustrates the expected pipeline outputs for three representative response scenarios, demonstrating the system's coherent behavior across the difficulty-correctness space.

TABLE III
EXPECTED PIPELINE OUTPUT FOR REPRESENTATIVE SCENARIOS

Scenario	Correct	Time (s)	Predicted Difficulty	Next Difficulty
Fast + Correct	Yes	10	Easy	Medium
Slow + Correct	Yes	50	Medium	Hard
Fast + Incorrect	No	10	Hard	Medium
Slow + Incorrect	No	55	Hard	Medium

The trace demonstrates that the system correctly escalates difficulty for students who answer quickly and correctly, reflecting mastery of easy material. Students who answer slowly but correctly are placed at Medium difficulty, reflecting effortful but successful performance. Students who answer incorrectly—regardless of speed—are classified as facing Hard questions and are stepped down, consistent with the adaptive rule set.

D. Advantages of LLM-Native Adaptation over Formula-Based Approaches

A key design decision in this system is the delegation of all difficulty reasoning to LLM agents rather than deterministic formulas. This choice offers several advantages. First, the adaptation rules can be expressed and modified entirely in natural language without rewriting scoring code, improving maintainability. Second, the agents can incorporate contextual nuance—such as recognizing borderline cases in time thresholds—that rigid numeric formulas cannot express. Third, the full reasoning chain is available in the agent output, providing educators with a natural language explanation for every difficulty decision, directly supporting the explainability goals outlined in the APOT framework [1]. Fourth, eliminating pre-calibrated IRT parameters makes the system deployable in any domain without prior item bank construction [4].

The primary trade-off is the non-determinism inherent in LLM outputs, which means identical inputs may occasionally produce slightly different outputs. This is mitigated by the structured output format enforced through task descriptions and the fallback parsing logic that defaults to "Medium" when output is ambiguous.

VII. CONCLUSION AND FUTURE WORK

This paper presented an Adaptive Multi-Agent MCQ Difficulty Prediction System that operationalizes the APOT framework of Outay *et al.* [1] in a fully LLM-driven, question-based adaptive testing context. By deploying three specialized CrewAI agents powered by Google Gemini 2.5 Flash—an MCQ Evaluator, a Difficulty Predictor, and an Adaptive MCQ Overseer—the system replaces deterministic scoring formulas and IRT pre-calibration with prompt-engineered LLM reasoning over answer correctness and response time.

The design-level analysis confirms that the three-agent pipeline provides complete coverage of the adaptive transition space, logically consistent behavior across representative performance scenarios, and a natural language audit trail for every difficulty decision. The system demonstrates that LLM-native multi-agent architectures can implement adaptive testing logic that is simultaneously interpretable, deployable without pre-calibrated item banks, and extensible through simple prompt modification.

Future work will focus on: (1) conducting user studies with diverse student populations to empirically validate the accuracy of LLM-predicted difficulty labels against expert annotations; (2) extending the system to multi-domain assessments ; (3) integrating session-level memory to enable cross-question performance tracking and long-horizon difficulty planning; (4) exploring prompt optimization techniques to reduce LLM output variability and improve structured output reliability; (5) comparing LLM-based adaptation decisions against IRT-based ground truth on standardized item banks to quantify agreement rates; and (6) developing a web-based deployment for concurrent multi-user assessment sessions with real-time difficulty dashboards for instructors.

REFERENCES

[1] F. Outay, N. Jabeur, F. Bellalouna, and T. Al Hamzi, "Multi-agent system-based framework for an intelligent management of competency building," *Smart Learning Environments*, vol. 11, no. 41, pp. 1–18, 2024, doi: 10.1186/s40561-024-00328-3.

[2] D. J. Weiss, "Computerized adaptive testing for effective and efficient measurement in counseling and education," *Measurement and Evaluation in Counseling and Development*, vol. 37, no. 2, pp. 70–84, 2004.

[3] F. M. Lord, *Applications of Item Response Theory to Practical Testing Problems*. Hillsdale, NJ, USA: Lawrence Erlbaum Associates, 1980.

[4] R. K. Hambleton and H. Swaminathan, *Item Response Theory: Principles and Applications*. Boston, MA, USA: Kluwer-Nijhoff, 1985.

[5] J. S. Liang *et al.*, "AutoGen: Enabling next-gen LLM applications via multi-agent conversation," *arXiv preprint arXiv:2308.08155*, 2023.

[6] C. Conati, A. Gertner, and K. VanLehn, "Using Bayesian networks to manage uncertainty in student modeling," *User Modeling and User-Adapted Interaction*, vol. 12, no. 4, pp. 371–417, 2002.

[7] M. Wooldridge, *An Introduction to MultiAgent Systems*, 2nd ed. Chichester, UK: John Wiley & Sons, 2009.
[8] B. S. Bloom *et al.*, *Taxonomy of Educational Objectives: The Classification of Educational Goals, Handbook I: Cognitive Domain*. New York, NY, USA: David McKay, 1956.
[9] L. S. Vygotsky, *Mind in Society: The Development of Higher Psychological Processes*. Cambridge, MA, USA: Harvard University Press, 1978.
[10] R. J. Mislevy, L. S. Steinberg, and R. G. Almond, "On the structure of educational assessments," *Measurement: Interdisciplinary Research and Perspectives*, vol. 1, no. 1, pp. 3–62, 2003.
[11] L. Chen *et al.*, "Large language models in education: A review of applications, opportunities, and challenges," *Educational Technology & Society*, vol. 26, no. 4, pp. 1–18, 2023.
[12] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA, USA: MIT Press, 2018.